Learning correspondence estimation for deformable 3D shapes

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Integrated Dual Degree (B.Tech + M.Tech)

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Computer Science and Engineering

by

Lubesh Kumar Sharma

(22CS30065)

Under the supervision of

Professor Ayan Chaudhury

Department of Computer Science and Engineering

Indian Institute of Technology Kharagpur

Autumn Semester, 2025-26

October 30, 2025

DECLARATION

I certify that

(a) The work contained in this report has been done by me under the guidance of my supervisor.

(b) The work has not been submitted to any other Institute for any degree or diploma.

(c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

(d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

Date: October 30, 2025 (Lubesh Kumar Sharma)
Place: Kharagpur (22CS30065)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

KHARAGPUR - 721302, INDIA

CERTIFICATE

This is to certify that the project report entitled "**Learning correspondence estimation for deformable 3D shapes**" submitted by **Lubesh Kumar Sharma** (Roll No. 22CS30065) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Integrated Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering is a record of bona fide work carried out by him under my supervision and guidance during Autumn Semester, 2025-26.

Date: October 30, 2025

Place: Kharagpur

Professor Ayan Chaudhury
Department of Computer Science and Engineering
Indian Institute of Technology Kharagpur
Kharagpur - 721302, India

Abstract

Name of the student: **Lubesh Kumar Sharma** Roll No: **22CS30065**

Degree for which submitted: **Integrated Dual Degree (B.Tech + M.Tech)**

Department: **Department of Computer Science and Engineering**

Thesis title: **Learning correspondence estimation for deformable 3D shapes**

Thesis supervisor: **Professor Ayan Chaudhury**

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Unsupervised point cloud shape correspondence seeks to establish dense, point-to-point mappings between deformable 3D shapes without manual annotations. However, this task is severely hampered by inherent challenges. Notably, the bilateral symmetry common in humans and the variability in object orientations frequently lead to catastrophic mispredictions of corresponding symmetrical parts. Concurrently, the inevitable presence of noise in point cloud data disrupts the learning of consistent geometric representations, further degrading correspondence accuracy. Moreover, features derived purely from local geometry lack the global context and semantic understanding necessary for robust matching across significant deformations. Lastly, prevalent unsupervised proxy tasks, such as reconstruction, often prove insufficient, failing to enforce strong geometric plausibility and suffering from issues like mode collapse.

To address these multifaceted challenges concurrently, this thesis proposes a novel,

integrated unsupervised framework. We tackle noise sensitivity and enhance feature robustness through a Self-Ensembling (student–teacher) architecture, leveraging consistency regularization between perturbed and clean data views. Crucially, before feature extraction, an Orientation-Aware Module explicitly aligns input point clouds, canonicalizing their poses to effectively mitigate errors arising from symmetry and varying orientations. To overcome the limitations of purely geometric features, our approach injects rich prior knowledge from pre-trained 2D vision models via a multi-view projection and feature aggregation scheme, providing essential global and semantic context. Finally, departing from weaker proxy tasks, our framework incorporates a novel Deformation-Based Learning Objective within the self-ensembling loop. This module uses the predicted correspondences to physically deform the source shape towards the target, grounding the feature learning process in geometric realism.

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Abbreviations

SE-ORNet	**S**elf-**E**nsembling **O**rientation-**A**ware **N**etwork
DV-Matcher	**D**eformation-based **N**on-**r**igid **P**oint **C**loud **M**atching
LG-NET	**L**ocal-and-**G**lobal attention **N**etwork
NeRF	**N**eural **R**adiance **F**ields
DPC	Unsupervised **D**eep **P**oint **C**orrespondence via Cross and Self Construction
DGCNN	**D**ynamic **G**raph **C**onvolutional **N**eural **N**etwork
OEM	**O**rientation **E**stimation **M**odule
FIM	**F**eature **I**nteraction **M**odule
EMA	**E**xponential **M**oving **A**verage
MLP	**M**ulti-**L**ayer **P**erceptron
CD	**C**hamfer **D**istance

Chapter 1

Introduction

1.1 The Fundamental Challenge of Non-Rigid Shape Correspondence

Estimating dense correspondences between 3D shapes is a cornerstone problem in computer vision and geometry processing. The goal is to establish a meaningful point-to-point mapping between two shapes, often represented as point clouds, that reflects their underlying semantic or structural similarity, even when they undergo significant non-rigid deformations. Achieving accurate correspondence is crucial for a multitude of applications, including human shape analysis (Anguelov et al., 2005), motion transfer (Sun et al., 2023), statistical modeling of anatomical variation (Adams and Elhabian, 2024), 3D reconstruction from dynamic scenes (Yu et al., 2018), and augmented reality.

While rigid alignment is well-studied, non-rigid correspondence presents profound challenges. Objects like human bodies can articulate and deform in complex ways, leading to large extrinsic variations in 3D coordinates between semantically identical points. Capturing this invariance requires methods that can reason about the intrinsic structure of shapes, often moving beyond simple spatial proximity.

1.2 Evolution Towards Unsupervised Learning on Point Clouds

Initial efforts in non-rigid correspondence often relied on structured mesh representations and sophisticated geometric techniques, such as spectral methods based on the Laplace-Beltrami operator (Ovsjanikov et al., 2012). While powerful, these methods typically require clean, manifold meshes, which are often unavailable in real-world scenarios where data is captured as raw, potentially noisy, and incomplete point clouds using sensors like LiDAR or depth cameras.

The advent of deep learning brought significant progress, particularly with methods capable of operating directly on unstructured point clouds, pioneered by architectures like PointNet (Qi et al., 2017a) and its successors like PointNet++ (Qi et al., 2017b) and DGCNN (Wang et al., 2019). Early deep learning approaches for correspondence often required supervised training (3D-CODED, (Groueix et al., 2018)), demanding vast amounts of meticulously annotated point-to-point mappings. However, the sheer cost and difficulty of generating such dense annotations severely limits the scalability and applicability of supervised methods.

This bottleneck motivated a critical shift towards *unsupervised* learning techniques. Landmark papers like CorrNet3D (Zeng et al., 2021) and DPC (Ginzburg and Raviv, 2021) introduced innovative self-supervised strategies. They proposed using feature similarity to induce a mapping and then training the feature extractor by minimizing a *reconstruction loss* – essentially, checking if one shape could be reconstructed as a weighted average of points from the other shape using the predicted mapping. This paradigm allowed models to learn meaningful correspondences directly from unlabeled point cloud data, representing a significant breakthrough.

1.3 Persistent Gaps in Unsupervised Correspondence

Despite the success of these foundational unsupervised methods, several persistent challenges remain, preventing robust and accurate performance across diverse and complex scenarios:

- **Orientation and Symmetry Ambiguity:** Many objects, particularly humans, exhibit bilateral symmetry. When shapes are presented in different orientations, methods relying solely on local geometric features often struggle to distinguish between symmetric counterparts (e.g., left vs. right limbs). This leads to geometrically inconsistent and semantically incorrect mappings, a critical failure point noted in subsequent analyses (Deng et al., 2023). Resolving this ambiguity is crucial for reliable correspondence.

- **Sensitivity to Noise and Data Imperfections:** Real-world point clouds are rarely pristine. Sensor noise, occlusions, varying point densities, and artifacts from preprocessing steps like downsampling can significantly perturb the input data. Feature extractors trained without explicit mechanisms for robustness often produce unstable representations, leading to degraded performance when applied outside clean, synthetic datasets (Deng et al., 2023). Methods need to be resilient to these imperfections.

- **Limited Semantic Understanding:** Features based purely on local point neighborhoods, while capturing geometry, often lack the broader semantic context needed to resolve ambiguities, especially under large deformations or when matching different instances of the same object class (style variations). Distinguishing between locally similar parts (e.g., knees and elbows) requires a higher-level understanding of the object's overall structure, motivating the integration of richer feature sources (Chen et al., 2025).

- **Weakness of Reconstruction-Based Objectives:** While reconstruction losses provide an effective unsupervised signal, they offer relatively weak geometric constraints. The process of averaging points does not guarantee a one-to-one mapping or enforce physical plausibility. This can lead to undesirable artifacts like "mode collapse," where the reconstructed shape becomes an indistinct average, providing a poor gradient signal for learning precise features (Chen et al., 2025). Stronger, more geometrically grounded objectives are desirable.

1.4 Thesis Contributions: An Integrated Robust Framework

Addressing these interrelated challenges—orientation/symmetry, noise, semantics, and geometric grounding—requires a holistic approach. This thesis presents a novel, unified framework for unsupervised non-rigid point cloud correspondence that integrates multiple complementary strategies into a single, cohesive pipeline, designed to operate directly on point clouds without requiring manual labels or mesh preprocessing.

Our primary contributions lie in the synergistic combination of techniques to overcome the aforementioned limitations:

1. **Self-Ensembling for Robustness:** To enhance resilience against **noise and perturbations**, we build our framework upon a student-teacher self-ensembling architecture (Tarvainen and Valpola, 2017). By enforcing consistency between a student network (trained on augmented data) and a stable teacher network, the learned feature representations gain significant robustness.

2. **Explicit Orientation Normalization:** We directly tackle **orientation and symmetry ambiguity** by incorporating an orientation estimation module early in the pipeline. This module predicts the relative rotation between shapes and aligns them to a consistent frame *before* feature extraction, effectively removing a major source of error for symmetric objects.

3. **Semantically Enriched Features via 2D Priors:** To provide necessary **semantic context**, we leverage the power of large-scale pre-trained 2D vision models. Using a multi-view projection and feature aggregation strategy (Zhang et al., 2023); (Chen et al., 2025), we inject rich visual features derived from 2D representations back into the 3D point features, complementing local geometry with global understanding.

4. **Deformation-Based Geometric Learning:** As a core contribution, we replace the **weak reconstruction proxy task** with a more powerful, deformation-based objective (Jiang et al., 2023), Non-rigid shape registration; (Chen et al., 2025). Integrated within the student branch, a dedicated deformation network uses the predicted correspondences to physically warp the source shape to align with the target. Minimizing the discrepancy between the deformed source and the target provides a strong, geometrically realistic supervisory signal, forcing the main feature extractor (an attention-based network, LG-NET) to learn correspondences that respect physical plausibility.

5. **Unified Training:** The student network is trained end-to-end to simultaneously satisfy feature-level consistency with the teacher (for robustness) and achieve accurate geometric alignment via the deformation loss (for geometric plausibility and accuracy).

By integrating these components, our framework aims to produce correspondences that are simultaneously accurate, robust to noise and orientation, semantically informed, and geometrically sound.

Chapter 2

Related Works

2.1 Learning Representations for Point Clouds

The effective application of deep learning to 3D point clouds necessitates architectures capable of handling the inherent irregularity and lack of order in point set data. Landmark work by PointNet (Qi et al., 2017a) introduced a foundational architecture that directly processes raw point coordinates, achieving permutation invariance through the use of symmetric functions, notably max-pooling, for feature aggregation. While transformative, PointNet's capacity to capture fine-grained local geometric structures proved limited. This limitation was subsequently addressed by PointNet++ (Qi et al., 2017b), which proposed a hierarchical approach. By recursively applying PointNet-like modules to nested partitions of the point set derived from metric space sampling and grouping, PointNet++ enabled the learning of local features at multiple contextual scales. Further exploration into local feature learning included attempts to generalize convolutional operations from image grids to point sets. PointCNN investigated learning an ($\mathcal{X}$)-transformation to weight and permute neighboring point features, aiming to establish a canonical ordering amenable to convolution. Concurrently, graph-based methods emerged as a powerful alternative. DGCNN (Wang et al., 2019) introduced the EdgeConv operator, which constructs k-nearest neighbor graphs dynamically in feature space at each layer and applies shared MLPs to edge features. This dynamic graph construction allows the network to adaptively group points based on learned semantics rather than fixed spatial proximity, proving effective for various point cloud tasks including correspondence (Ginzburg and Raviv, 2021), (Deng et al., 2023). Architectural elements inspired by such local and global aggregation strategies, including attention mechanisms operating over feature-defined neighborhoods, are utilized in the feature extraction network employed within this thesis (Chen et al., 2025).

2.2 Non-Rigid Shape Correspondence

Establishing correspondences between non-rigidly deforming shapes remains a central problem in geometry processing. A significant body of work operates on mesh representations, often leveraging spectral techniques. The functional maps framework (Ovsjanikov et al., 2012) provides an influential paradigm, representing shape mappings implicitly as linear operators between function spaces spanned by basis functions, commonly the Laplace-Beltrami operator eigenfunctions. This formulation offers advantages in compactness and the ability to incorporate constraints algebraically. Deep learning has been integrated into this framework to learn more effective basis functions or map representations (e.g., Litany et al., 2017, FMNet). Nonetheless, the reliance on mesh structures and spectral computations poses challenges for direct application to unstructured, potentially noisy point clouds commonly encountered in practice.

Early deep learning approaches targeting shape correspondence often necessitated supervised training signals. For instance, 3D-CODED (Groueix et al., 2018) utilized learned elementary 3D structures and their deformations to establish correspondences but relied on ground-truth point pairings during training. The significant expense and practical difficulty of acquiring dense correspondence annotations for large and diverse 3D datasets underscore the critical need for unsupervised methodologies.

2.3 Unsupervised Point Cloud Correspondence

The pursuit of unsupervised correspondence learning directly from point clouds has led to innovative self-supervised strategies. CorrNet3D (Zeng et al., 2021) pioneered an end-to-end framework based on a reconstruction objective. Features learned for source and target point clouds induce a soft correspondence map, which is then used to reconstruct one shape from the other; minimizing the reconstruction error serves as the training signal. DPC (Ginzburg and Raviv, 2021) refined this concept, introducing specific cross- and self-construction losses to enhance the quality and continuity of the learned point representations, establishing reconstruction-based learning as a strong baseline.

However, subsequent research identified critical limitations in these approaches. SE-ORNet (Deng et al., 2023) highlighted the vulnerability of reconstruction-based methods to ambiguities arising from object symmetry and varying input orientations, often resulting in incorrect symmetric part matching. It also noted the lack of inherent robustness to noise. SE-ORNet addressed these via an explicit Orientation Estimation Module for canonical alignment and a self-ensembling framework employing consistency losses to improve noise resilience. Independently, DV-Matcher (Chen et al., 2025) identified the semantic limitations of purely geometric features and the geometric weakness of the reconstruction proxy task itself, which can lead to mode collapse. DV-Matcher proposed integrating semantic priors from pre-trained 2D vision models via visual encoding and introduced a novel, geometrically grounded deformation-based learning objective, implemented with the attention-based LG-NET (Chen et al., 2025) architecture, where correspondence quality is measured by the success of an induced physical alignment.

2.4 Summary

The preceding review delineates the evolution of non-rigid shape correspondence, particularly highlighting the trajectory towards unsupervised learning directly from point cloud data. Foundational architectures like PointNet (Qi et al., 2017a), PointNet++ (Qi et al., 2017b), and DGCNN (Wang et al., 2019) enabled deep learning on point sets, while methods such as CorrNet3D (Zeng et al., 2021) and DPC (Ginzburg and Raviv, 2021) established the viability of reconstruction-based self-supervision. However, the literature also reveals persistent challenges inherent to this domain: ambiguity arising from object symmetries and varying orientations, sensitivity to noise and data imperfections prevalent in raw scans, the limitations of purely geometric features in capturing necessary semantic context, and the potential geometric weaknesses of reconstruction as a proxy task. Recent advancements have proposed targeted solutions, such as explicit orientation handling SE-ORNet (Deng et al., 2023), techniques for enhancing feature robustness like self-ensembling (Tarvainen and Valpola, 2017), methods for incorporating richer semantic information, potentially from external knowledge sources like pre-trained vision models DV-Matcher (Chen et al., 2025), and the exploration of alternative, geometrically stronger self-supervised objectives like deformation (Chen et al., 2025).

2.5 Project Scope and Approach

This project builds upon the reviewed advancements in unsupervised point cloud correspondence by implementing and evaluating a framework that integrates several contemporary techniques. Specifically, we investigate a pipeline combining an orientation alignment module, a self-ensembling (student-teacher) architecture for robustness, a visual encoding scheme leveraging pre-trained 2D models for semantic feature enrichment, and a deformation-based objective function for geometrically-aware learning, utilizing the attention-based LG-NET (Chen et al., 2025) feature extractor. The focus of this work lies in the practical realization of this combined methodology and its empirical assessment on standard human shape correspondence benchmarks, operating directly on point cloud data without manual supervision.

Chapter 3

Methodology

The goal of this work is to establish dense, point-to-point correspondences between non-rigidly deforming 3D shapes, represented as a source point cloud $X = \{x_i\}_{i=1}^{N} \subset \mathbb{R}^3$ and a target point cloud $Y = \{y_j\}_{j=1}^{M} \subset \mathbb{R}^3$, within an unsupervised learning framework. The architecture operates directly on point clouds without requiring manual annotations or mesh structures.

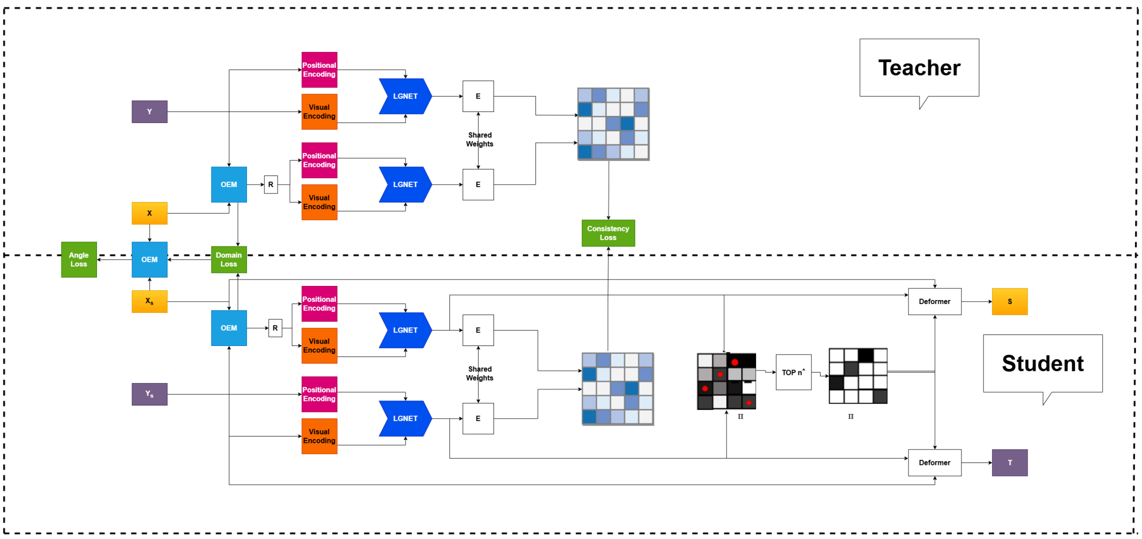

FIGURE 3.1: The overview of our Learning correspondence estimation for deformable 3D shapes.

3.1 Overview

The core of our method is a Siamese, student-teacher architecture based on the Mean Teacher paradigm (Tarvainen and Valpola, 2017), designed to enhance robustness to noise and other perturbations. Input point cloud pairs are processed through two parallel streams: a teacher branch receiving weakly augmented ("clean") data (X, Y), and a student branch receiving strongly augmented data (X_s, Y_s) involving significant noise and random rotations.

A key initial step involves passing both pairs through an Orientation Estimation Module (OEM) inspired by SE-ORNet (Deng et al., 2023). Trained using adversarial domain adaptation, the OEM predicts and corrects relative orientations, aligning the shapes $(\tilde{X}, Y)$ and $(\tilde{X}_s, Y_s)$ to a canonical frame. This pre-alignment mitigates errors arising from bilateral symmetry and rotational variance.

Following alignment, point features are enriched using a visual encoding process (Chen et al., 2025) that incorporates semantic information derived from pre-trained 2D vision models. These enhanced features, combined with positional encodings, are fed into the main feature extraction backbone – the LG-NET (Chen et al., 2025) architecture – implemented within both the student and teacher branches (with shared architecture but separate weights linked via EMA).

The student network's learned features $F_{\tilde{X}_s}, F_{Y_s}$ are utilized in two ways. Firstly, they are compared against the teacher's features $F_{\tilde{X}}, F_Y$ via consistency losses, as proposed in SE-ORNet (Deng et al., 2023), to enforce robustness. Secondly, they drive a deformation module (Chen et al., 2025) that physically warps the source shape $\tilde{X}_s$ towards the target Y_s. The quality of this deformation provides the primary geometric supervision for the student network.

The student network parameters are trained end-to-end by minimizing a composite loss function comprising the orientation supervision losses $(\mathcal{L}_{angle}, \mathcal{L}_{domain})$, the feature consistency losses $(\mathcal{L}_{ccs}, \mathcal{L}_{css})$, and the suite of geometric losses from the deformation module ($\mathcal{L}_{deform}, \mathcal{L}_{arap}, \mathcal{L}_{smooth}, \mathcal{L}_{geo}$). Teacher parameters are updated via EMA. Inference is performed using the trained student network.

3.2 Data Augmentation and Orientation Alignment

For a raw input pair (X_{raw}, Y_{raw}), the teacher receives a minimally perturbed version (X, Y), while the student receives a strongly augmented version (X_s, Y_s) where the source X_s includes significant Gaussian noise and a random rotation $\mathcal{R}(\theta_x)$ with a known angle label θ_x.

Both pairs are processed by the Orientation Estimation Module (OEM) (Deng et al., 2023). This module employs a feature encoder, a Feature Interaction Module (FIM) for inter-shape feature fusion via graph convolution, and prediction heads for rotation classification and domain discrimination. The OEM is trained on the student inputs using the angle loss $\mathcal{L}_{angle}$ (cross-entropy comparing predicted rotation bin probabilities $\hat{\theta}_{x_s}$ with the ground-truth label θ_x) and the domain loss $\mathcal{L}_{domain}$ (Focal Loss on the discriminator's output d assessing if input is real or augmented), facilitated by a Gradient Reversal Layer (GRL) for adversarial training:

$$\mathcal{L}_{angle} = -\sum_{m=1}^{M} \theta_x^{(m)} \log \hat{\theta}_{x_s}^{(m)} \tag{3.1}$$

$$\mathcal{L}_{domain} = -(1-d)^{\gamma} \log d \tag{3.2}$$

The predicted rotation $\hat{\mathcal{R}}$ from the OEM is applied to the source cloud in both branches, yielding orientation-aligned pairs $(\tilde{X}, Y)$ and $(\tilde{X}_s, Y_s)$ for subsequent processing.

3.3 Visually-Informed Feature Encoding

To incorporate semantic information, we apply the visual encoding method (Chen et al., 2025) to the aligned point clouds $P \in \{\tilde{X}, Y, \tilde{X}_s, Y_s\}$. This involves:

1. **Multi-view Projection:** Projecting P onto xy, yz, and xz planes to create depth-encoded images $I^{(v)}$, with hole-filling and pseudo-coloring.
2. **2D Feature Extraction:** Using a pre-trained vision model such as DINOv2 (Oquab et al., 2024) with feature upsampling through FeatUp (Fu et al., 2024) to extract per-pixel features $F_{img}^{(v)}$.
3. **Feature Lifting:** Mapping pixel features back to 3D points $F_{pt}^{(v)}$.
4. **Concatenation:** Combining features across views to form $F_{vis}(P) \in \mathbb{R}^{N \times 3C}$.

This visual encoding $F_{vis}(P)$ is fused with positional encoding $\gamma(P)$ as in NeRF (Mildenhall et al., 2021) via an LBR projection and summation to produce the input $F_{in}(P)$ for the LG-NET (Chen et al., 2025) backbone.

3.4 Siamese LG-NET Backbone

Feature extraction is performed using the LG-NET (Chen et al., 2025) architecture (as suggested in DV-Matcher (Chen et al., 2025)) within the Siamese student (f_θ) and teacher ($f_{\tilde{\theta}}$) networks. LG-NET processes $F_{in}(P)$ using parallel global and local attention pathways, where local neighborhoods are determined dynamically in feature space. Features from both pathways are fused to produce the final point embeddings $F_P \in \mathbb{R}^{N \times C'}$. The student outputs features $F_{\tilde{X}_s}, F_{Y_s}$, and the teacher outputs $F_{\tilde{X}}, F_Y$.

3.5 Correspondence Map and Self-Ensembling Consistency

Dense similarity matrices $S_{\tilde{X}Y}$ (teacher) and $S_{\tilde{X}_s Y_s}$ (student) are computed via cosine similarity between the output features. Self-similarity matrices $S_{\tilde{X}\tilde{X}}$ and $S_{\tilde{X}_s\tilde{X}_s}$ are also computed.

Robustness is enforced via the consistency losses defined in SE-ORNet (Deng et al., 2023), comparing student and teacher similarities:

$$\mathcal{L}_{ccs} = \text{SmoothL1}(S_{\tilde{X}Y}, S_{\tilde{X}_s Y_s}) \tag{3.3}$$

$$\mathcal{L}_{css} = \text{SmoothL1}(S_{\tilde{X}\tilde{X}}, S_{\tilde{X}_s\tilde{X}_s}) \tag{3.4}$$

For the deformation module, a soft correspondence map $\hat{\Pi}_{\tilde{X}_s Y_s}$ is derived from the student's similarity matrix $S_{\tilde{X}_s Y_s}$, typically involving softmax normalization and optional top-k sparsification, as in DV-Matcher (Chen et al., 2025).

3.6 Deformation-Based Geometric Objective (Student Branch)

The geometric learning signal is provided by the deformation module (Chen et al., 2025), applied solely to the student branch outputs. Based on the correspondence $\hat{\Pi}_{\tilde{X}_s Y_s}$, it deforms the source $\tilde{X}_s$ to align with the target Y_s, yielding $\hat{X}_s$. This process involves predicting rigid transformations $\{R_k, t_k\}$ for deformation graph nodes $\tilde{X}_{s,D}$ using an MLP conditioned on node features and pulled-back target features, followed by warping the full point cloud.

This deformation is supervised using the four geometric losses proposed in DV-Matcher (Chen et al., 2025), applied symmetrically for $\tilde{X}_s \leftrightarrow Y_s$:

- **Deformation Loss ($\mathcal{L}_{deform}$):** Chamfer Distance $CD(\hat{X}_s, Y_s)$ measuring alignment quality.
- **ARAP Loss ($\mathcal{L}_{arap}$):** Regularizes the rigidity of the deformation graph transformations.
- **Smoothness Loss ($\mathcal{L}_{smooth}$):** Encourages neighborhood preservation by the soft map, $CD(Y_s, \hat{\Pi}_{\tilde{X}_s Y_s} Y_s)$.
- **Geometric Similarity Loss ($\mathcal{L}_{geo}$):** Aligns embedding space distances with surface geodesic distances for local neighborhoods.

3.7 Total Loss Function

The student network parameters θ (encompassing the OEM encoder, LG-NET (Chen et al., 2025), and Deformer) are updated by minimizing a total loss, which combines the objectives described in the previous sections. This composite loss includes the orientation-related losses ($\mathcal{L}_{angle}$, $\mathcal{L}_{domain}$) and the feature consistency losses ($\mathcal{L}_{ccs}$, $\mathcal{L}_{css}$) adapted from SE-ORNet (Deng et al., 2023). These are combined with the suite of geometric losses introduced in DV-Matcher (Chen et al., 2025), which includes the deformation loss ($\mathcal{L}_{deform}$), the rigidity regularizer ($\mathcal{L}_{arap}$), the map smoothness loss ($\mathcal{L}_{smooth}$), and the geometric similarity loss ($\mathcal{L}_{geo}$).

The total loss is summed over both mapping directions:

$$\mathcal{L}_{total} = \mathcal{L}_{total}^{(\tilde{X}_s \to Y_s)} + \mathcal{L}_{total}^{(Y_s \to \tilde{X}_s)} \tag{3.5}$$

where the loss for a single direction is a weighted summation of all components:

$$\begin{aligned} \mathcal{L}_{total}^{(\to)} = \lambda_{angle}\mathcal{L}_{angle} \quad & + \lambda_{domain}\mathcal{L}_{domain} + \lambda_{ccs}\mathcal{L}_{ccs} \quad & + \lambda_{css}\mathcal{L}_{css} \\ + \lambda_{deform}\mathcal{L}_{deform} \quad & + \lambda_{arap}\mathcal{L}_{arap} + \lambda_{smooth}\mathcal{L}_{smooth} \quad & + \lambda_{geo}\mathcal{L}_{geo} \end{aligned} \tag{3.6}$$

The λ values serve as hyperparameters balancing the contributions of the different terms.

3.8 Training Strategy and Inference

Training proceeds by updating student parameters θ via gradient descent on $\mathcal{L}_{total}$, while teacher parameters $\tilde{\theta}$ are updated via EMA:

$$\tilde{\theta} \leftarrow \beta\tilde{\theta} + (1-\beta)\theta \tag{3.7}$$

For inference, given a new pair (X_{new}, Y_{new}), the trained student network f_θ (including OEM alignment, visual encoding, and LG-NET (Chen et al., 2025)) is used to compute final features $F_{X_{new}}, F_{Y_{new}}$. Correspondences are then established via nearest neighbor search in the feature space based on cosine similarity.

Chapter 4

Experiments and Results

To evaluate the performance of our integrated unsupervised correspondence framework, we conduct a series of experiments on standard non-rigid 3D human shape benchmarks. This chapter details the experimental setup, including the datasets, evaluation metrics, and baseline methods used for comparison. We then present both qualitative and quantitative results, followed by a discussion of the findings.

4.1 Experimental Setup

Datasets: Our evaluation is performed on three widely-used public datasets for human shape correspondence:

1. **SCAPE_r**: A standard benchmark dataset consisting of 3D human models in a variety of poses. Following the experimental setup of **DV-Matcher** (Chen et al., 2025), we evaluate our method on a *partial* version of this dataset, testing the model's ability to handle partial-to-full shape matching.
2. **SHREC_19**: A benchmark for non-rigid human shape matching. We utilize the standard test pairs provided by this dataset.

3. **FAUST**: A dataset of 3D human scans, also a standard benchmark for correspondence tasks.

Baselines: We compare our proposed method ("Ours") against two primary state-of-the-art frameworks from which its core components are derived:

- **SE-ORNet (CVPR 2023)** (Deng et al., 2023): An unsupervised method based on self-ensembling and an Orientation Estimation Module (OEM) to handle noise and symmetry.
- **DV-Matcher (CVPR 2025)** (Chen et al., 2025): An unsupervised method utilizing visual encoding (from 2D pre-trained models) and a deformation-based learning objective.

Evaluation Metric: We measure correspondence quality using the **mean geodesic error**. For a given source point, the geodesic error is the shortest distance *along the surface* of the target mesh from the ground-truth corresponding point to the point predicted by our method. This metric is more accurate for deformable shapes than simple Euclidean distance as it respects the intrinsic structure of the surface. The mean geodesic error is the average of this error over all points in the test set. We report this error as a percentage (multiplied by 100) for clarity in our quantitative results.

4.2 Qualitative Results

Visualizing the correspondence map provides an intuitive understanding of a method's performance. In Figure 4.1 we present representative qualitative comparisons between our method, **SE-ORNet** (Deng et al., 2023), and **DV-Matcher** (Chen et al., 2025) across the **SCAPE_r**, **SHREC_19**, and **FAUST** datasets. The correspondences are visualized by transferring color from a source shape to the target shape

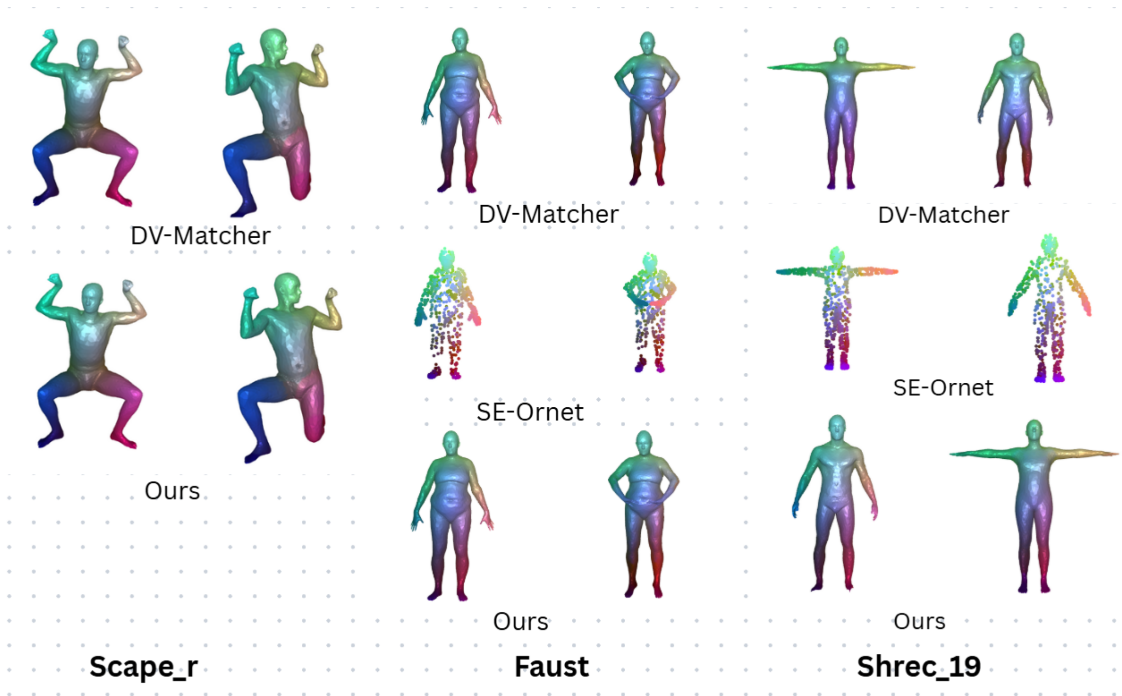

FIGURE 4.1: Visual correspondence results on SCAPE_r, Faust, SHREC_19.

based on the predicted point-to-point mapping. These visualizations allow for a qualitative assessment of the geometric plausibility and semantic correctness of the correspondences, particularly in challenging regions such as limbs and digits.

4.3 Quantitative Results

We present a quantitative comparison of mean geodesic error in Table 4.1. The performance of our integrated framework is benchmarked against the baseline methods **SE-ORNet (CVPR 2023)** and **DV-Matcher (CVPR 2025)**.

Method	**SCAPE_r (%)**	**SHREC'19 (%)**	**FAUST (%)**
SE-ORNet (CVPR 2023)	N/A	26.29	47.89
DV-Matcher (CVPR 2025)	10.14	35.57	11.10
Ours	**10.19**	**34.94**	**23.99**

TABLE 4.1: Quantitative comparison of mean geodesic error (in %), lower is better, on human correspondence benchmarks.

**Indicates result reported on downsampled data, as per original paper's experiment.*

References for Methods:

- SE-ORNet (CVPR 2023): (Deng et al., 2023)
- DV-Matcher (CVPR 2025): (Chen et al., 2025)

4.4 Discussion of Results

The quantitative results presented in Table 4.1 provide key insights into the performance of our integrated framework.

On the **SCAPE_r** partial dataset, our method (10.19%) achieves performance that is highly comparable to **DV-Matcher** (Chen et al., 2025) (10.14%). This suggests that the integration of the self-ensembling and orientation alignment modules did not detract from the strong geometric and semantic capabilities of the deformation-based framework in this partial matching context.

For the **SHREC'19** dataset, our approach (34.94%) shows a slight improvement over **DV-Matcher** (Chen et al., 2025) (35.57%). It is noteworthy that **SE-ORNet**

(Deng et al., 2023) (26.29%) performs well on this benchmark, which may be attributed to differences in the experimental setup (e.g., partial vs. full data) or the dataset's specific characteristics, such as rotational ambiguity, which the OEM explicitly targets.

On the **FAUST** dataset, our result (23.99%) lies between the strong performance of **DV-Matcher** (Chen et al., 2025) (11.10%) and the higher error reported for the downsampled **SE-ORNet** (Deng et al., 2023) experiment (47.89%). This variance indicates that the complex interplay between multiple loss functions (orientation, consistency, and deformation) and network components in our unified model may require further dataset-specific tuning to achieve optimal performance consistently.

In summary, the results demonstrate that our integrated framework produces competitive and semantically plausible correspondences, particularly showing strong comparative performance on the **SCAPE_r** and **SHREC'19** datasets. The performance variation on **FAUST** suggests that balancing the contributions of different modules remains a non-trivial task and presents a clear direction for future improvement.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This thesis investigated the challenging problem of unsupervised dense correspondence estimation between non-rigidly deforming 3D point clouds. Current methodologies face significant hurdles, including ambiguity arising from object symmetry and varying orientations, sensitivity to noise inherent in sensor data, limitations of purely geometric features in capturing essential semantic context, and the geometric weaknesses associated with common self-supervised proxy tasks like reconstruction. Addressing these issues concurrently is critical for advancing the field towards more robust and practical solutions.

In response, this work implemented and evaluated an integrated unsupervised framework designed to mitigate these challenges in a unified manner. The developed pipeline incorporates several key strategies: explicit orientation alignment via a dedicated module to canonicalize poses and resolve symmetry issues; a self-ensembling (student-teacher) architecture employing consistency regularization to enhance robustness against noise and data perturbations; feature enrichment through a visual encoding scheme that integrates semantic information from pre-trained 2D vision models; and a deformation-based learning objective, utilizing an attention-based feature extractor (LG-NET (Chen et al., 2025)), to provide a geometrically strong supervisory signal grounded in physical alignment.

The framework was trained end-to-end directly on point cloud data without manual correspondence labels. Experimental evaluations were conducted on standard human shape benchmark datasets, including **SCAPE_r**, **SHREC'19**, **FAUST_r**. The analysis focused on assessing the integration of the different components and evaluating the resulting correspondence quality relative to established unsupervised baselines.

The results offer insights into the complexities and potential benefits of combining these advanced techniques. While the integrated approach demonstrates promise in leveraging complementary strengths for robust, geometrically aware, and semantically informed correspondence, the findings also underscore the need for careful calibration and further refinement to fully realize the potential of such hybrid systems. This thesis contributes a practical implementation and initial analysis of a unified strategy for tackling multiple key challenges in unsupervised correspondence estimation.

5.2 Future Work

The framework and analysis presented herein suggest several promising directions for future research and development:

1. **Systematic Optimization and Parameter Tuning:** Further improvements in correspondence accuracy could likely be achieved through rigorous optimization of the existing pipeline. This includes exploring different network

hyperparameters, loss weightings, optimizer configurations, and potentially architectural variations within the feature extraction and deformation modules to enhance performance on benchmark datasets.

2. **Advanced Loss Formulations:** Investigating alternative or supplementary loss functions could further enhance specific properties of the learned correspondences. This might involve exploring different metrics for feature consistency within the self-ensembling framework, incorporating regularizers that explicitly promote map bijectivity or cycle consistency, or developing more sophisticated geometric constraints for the deformation objective.

3. **Dedicated Partial Matching Mechanisms:** To improve performance on incomplete data, dedicated mechanisms for handling partiality could be integrated. Potential approaches include incorporating explicit overlap prediction, utilizing partial-aware distance metrics within the deformation loss, or designing feature extractors that are inherently more robust to missing geometric information.

4. **Scalability and Computational Efficiency:** Enhancing the framework's scalability to significantly larger datasets (e.g., hundreds of thousands of shapes) and improving computational efficiency are crucial for practical applications. This could involve optimizing computationally intensive steps like visual feature extraction or deformation graph processing, investigating model parallelism, or exploring more efficient network architectures. Reducing inference time is also a key consideration.

5. **Robustness to Arbitrary Orientations:** Extending the system's robustness beyond alignment around a primary axis to handle arbitrary initial 3D rotations would significantly broaden its applicability. This might require developing more sophisticated orientation prediction modules or leveraging feature representations inherently invariant to general rotations.

6. **Generalization Across Domains:** Thoroughly evaluating and potentially adapting the framework for robust performance on diverse real-world datasets, including scans with significant noise and clutter, as well as extending it to different categories of deformable objects beyond human shapes (e.g., animals, garments, medical data), would be essential for demonstrating its broad utility.

Appendix A

Technical Details of Network Components

A.1 Orientation Estimation Module (OEM)

The Orientation Estimation Module (OEM)(Deng et al., 2023), is responsible for predicting the relative rotation between a pair of input point clouds and enabling their alignment to a canonical orientation prior to correspondence estimation. This step is crucial for mitigating errors caused by bilateral symmetry and varying initial poses.

The OEM processes a pair of point clouds, source P^s and target P^t, represented by their initial features (e.g., coordinates or outputs from a preliminary feature encoder). Its architecture comprises:

1. **Feature Encoder:** A lightweight point cloud feature extractor (e.g., a simplified DGCNN (Wang et al., 2019) processes P^s and P^t independently to obtain initial point-wise features F^s_{in} and F^t_{in}.

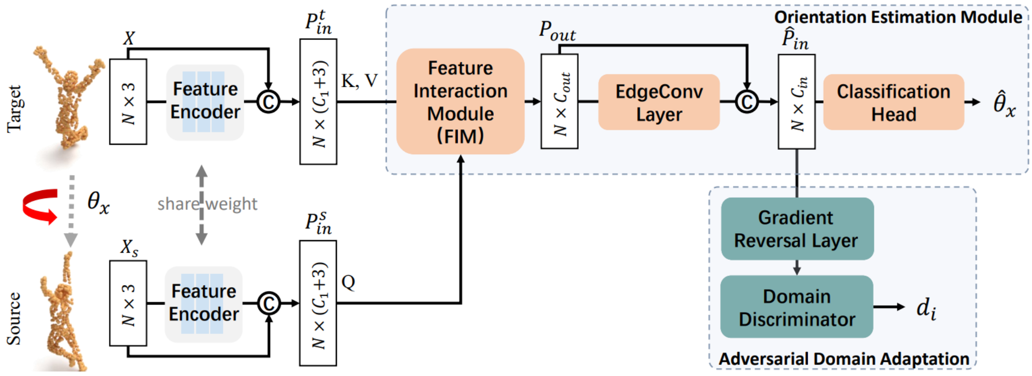

FIGURE A.1: The overview of Orientation Estimation. Source: Deng et al., SE-ORNet (2023). © 2023.

2. **Feature Interaction Module (FIM):** This module facilitates information exchange between the source and target features, producing fused features F_{out}.

3. **Feature Refinement:** The fused features F_{out} are refined, for example, using an additional EdgeConv layer (DGCNN (Wang et al., 2019)), yielding enhanced features $\hat{F}_{in}$.

4. **Rotation Classification Head:** Inspired by orientation prediction in object detection (SE-ORNet (Deng et al., 2023)), this head treats rotation estimation as a classification task. It aggregates enhanced features $\hat{F}_{in}$ into a global descriptor and passes it through an MLP to predict a probability distribution $\hat{\theta}$ over M discrete angle bins covering the rotation range (e.g., 360° around the vertical axis). The bin with the highest probability determines the predicted rotation $\hat{\mathcal{R}}$.

5. **Adversarial Domain Adaptation Head:** To enable training using known rotations from augmented data while generalizing to real data with unknown rotations, an adversarial discriminator is employed. This head (e.g., a PointNet-like MLP) takes the features $\hat{F}_{in}$ and predicts the probability d that they

originated from a 'real' (weakly augmented) pair versus a 'fake' (strongly augmented, synthetically rotated) pair. It is trained adversarially using a Gradient Reversal Layer (GRL).

The OEM is trained using the Angle Loss $\mathcal{L}_{angle}$ and Domain Loss $\mathcal{L}_{domain}$ supervising the rotation and discriminator heads respectively.

A.2 Feature Interaction Module (FIM)

The Feature Interaction Module (FIM), part of the OEM (Deng et al., 2023), performs cross-attention between the source and target point clouds to generate features informed by the geometry of the counterpart shape.

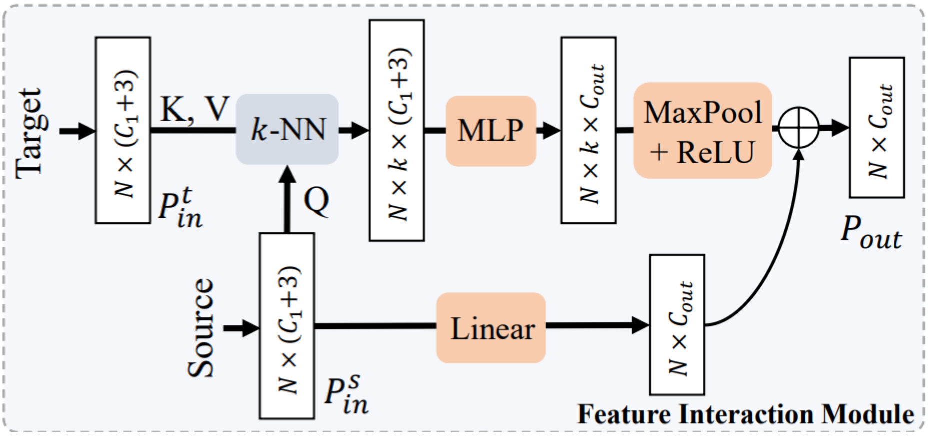

FIGURE A.2: Feature Interaction Module. Source: Deng et al., SE-ORNet (2023). © 2023.

Let the input features be $F^s_{in} \in \mathbb{R}^{N \times C_1}$ and $F^t_{in} \in \mathbb{R}^{M \times C_1}$. The FIM operates as follows:

1. **Neighbor Search:** For each point p^s_i in the source, identify k-nearest neighbors $\{p^t_{i,j}\}^k_{j=1}$ in the target based on Euclidean distance.

2. **Edge Feature Computation:** For each pair $(p_i^s, p_{i,j}^t)$, compute an edge feature capturing both feature and spatial relationships, e.g., $e_{ij} = MLP(p_i^s, p_{i,j}^t - p_i^s)$.

3. **Aggregation:** Aggregate the k edge features using max-pooling followed by activation (ReLU), producing $f_{out,i}^s$.

4. **Residual Connection:** Combine F_{out}^s with F_{in}^s (via addition or concatenation) to facilitate gradient flow.

This mechanism allows source features to be influenced by contextually relevant target features.

A.3 Visual Encoding

The Visual Encoding module (Chen et al., 2025), enriches raw 3D point coordinates with semantic information from pre-trained 2D vision models.

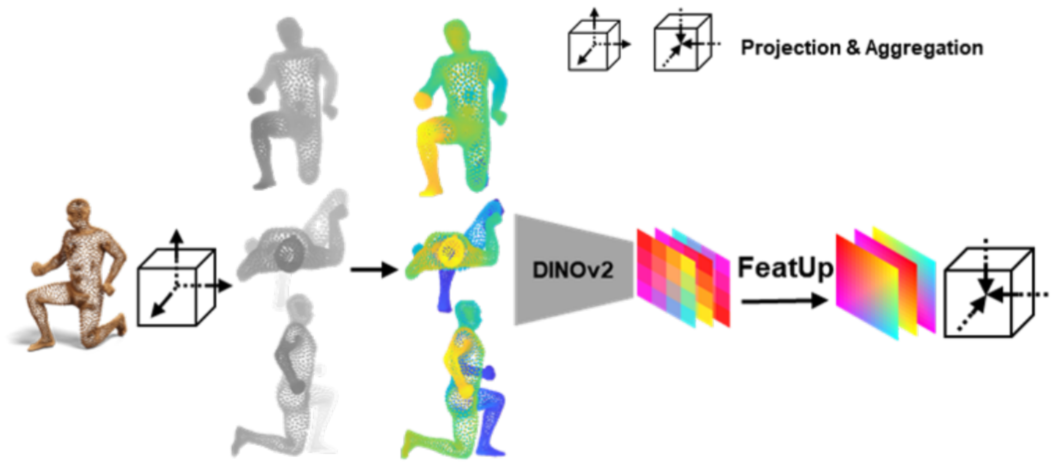

FIGURE A.3: The schematic illustration of the proposed Visual Encoding. Source: Chen et al., DV-Matcher (2025). © 2025.

The process includes:

1. **Multi-view Projection:** Project the 3D point cloud onto orthogonal 2D planes (xy, yz, xz). Pixel intensity encodes depth via $I(u_i, v_i) = \text{sigmoid}(z_i)$.

2. **Image Post-processing:** Apply hole filling and pseudo-color mapping (e.g., PiYG colormap).

3. **2D Feature Extraction & Upsampling:** Process each image via a frozen 2D vision backbone (e.g., DINOv2 (Oquab et al., 2024)) followed by a feature upsampling network (FeatUp (Fu et al., 2024)).

4. **Feature Lifting:** Map dense 2D features back to 3D points based on their projections, producing $F_{pt}^{(v)}$.

5. **Concatenation:** Combine all view features to form $F_{vis}(P) = [F_{pt}^{(xy)}, F_{pt}^{(yz)}, F_{pt}^{(xz)}] \in \mathbb{R}^{N\times 3C}$.

This representation provides semantically rich per-point features complementary to geometric encodings.

A.4 Local and Global Attention Network (LG-NET)

The LG-NET architecture (Chen et al., 2025), serves as the main feature extraction backbone. It fuses visual and positional encodings to generate final discriminative embeddings.

1. **Input Fusion:** Input $F_{in}(P) = LBR(F_{vis}(P)) + \gamma(P)$, where γ denotes positional encoding.

2. **Dual-Pathway Architecture:**

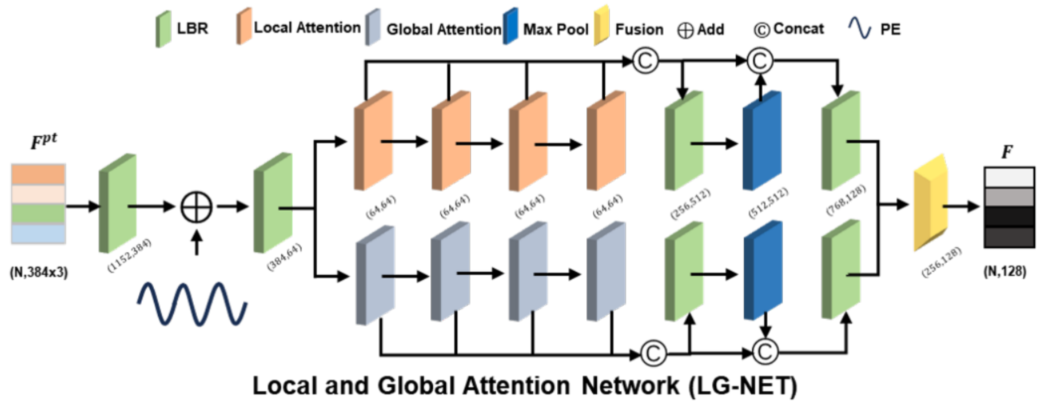

FIGURE A.4: The schematic illustration of the LG-NET. Source: Chen et al., DV-Matcher (2025). © 2025.

- *Global Attention Path:* Employs stacked self-attention blocks (similar to point transformer (Guo et al., 2024)) for global context modeling.
- *Local Attention Path:* Uses local attention within dynamically updated k-NN neighborhoods (following DGCNN (Wang et al., 2019))

3. **Fusion Module:** Merges outputs via concatenation and refinement layers (e.g., stacked N2P attention blocks (Chen et al., 2025)), yielding $F_P \in \mathbb{R}^{N\times C'}$.

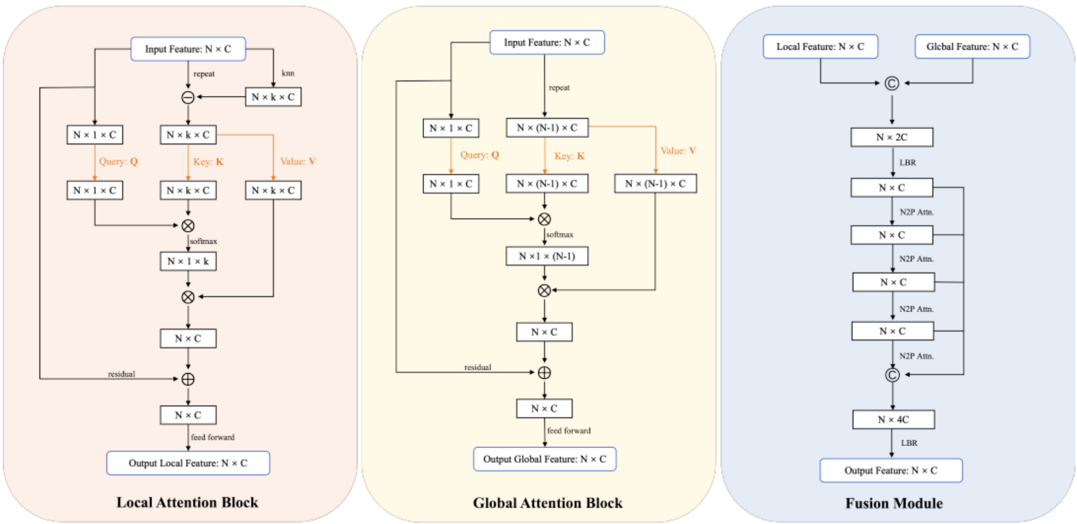

FIGURE A.5: The schematic illustration of the main blocks of LG-NET. Source: Chen et al., DV-Matcher (2025). © 2025.

A.5 Deformer Module

The Deformer module (Chen et al., 2025), implements the deformation-based learning objective. Given a source P^s and target P^t, along with the predicted correspondence map $\hat{\Pi}_{st}$, it predicts non-rigid deformations to warp P^s toward P^t.

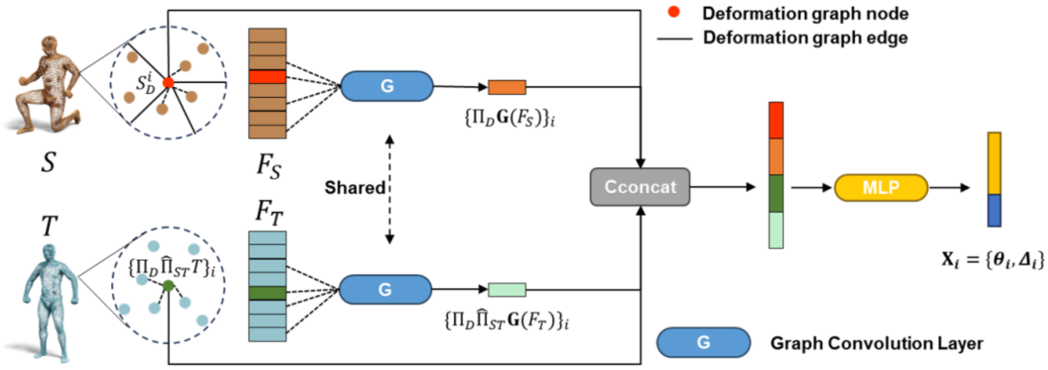

FIGURE A.6: The schematic illustration of the Deformer. Source: Chen et al., DV-Matcher (2025). © 2025.

1. **Deformation Graph Construction:** Sample K graph nodes P_D^s from P^s using Farthest Point Sampling (FPS).

2. **Transformation Prediction:** For each node $p_{D,k}^s$, predict rigid transformations (R_k, t_k) using

$$\{(R_k, t_k)\}_{k=1}^{K} = MLP(P_D^s, F_D^s, \hat{P}_D^t, \hat{F}_D^t)$$

where rotations R_k use the 6D rotation representation (Zhou et al., 2019).

3. **Point Cloud Warping:** Propagate node transformations to all points via smooth weighting (e.g., embedded deformation (Sumner et al., 2007)), yielding $\hat{P}^s = \mathcal{DG}(P^s, \{R_k, t_k\})$.

The deformation quality relative to P^t (measured by $\mathcal{L}_{deform}$) provides geometric supervision, regularized by the ARAP loss $\mathcal{L}_{arap}$ applied to the node transformations.

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